

Competitive Programming

Week 1

Simon Fraser University



2025 World Finals in Baku



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Following ...



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The School of Computing Science is proud to have hosted the 2025 Pacific Northwest Regional International Collegiate Programming Contest ([@icpcnews](#)) last weekend on Burnaby campus!

Held at our Computing Science Instructional Labs (CSIL), top students from [@simonfraseru](#), [@universityofbc](#), [@ubcokanagan](#), [@universityofvictoria](#), and [@trinitywestern](#) gathered to compete in algorithmic challenges that tested their teamwork, creativity, and problem-solving skills.

Congratulations to all the participants for their amazing performance and resilience throughout the competition.

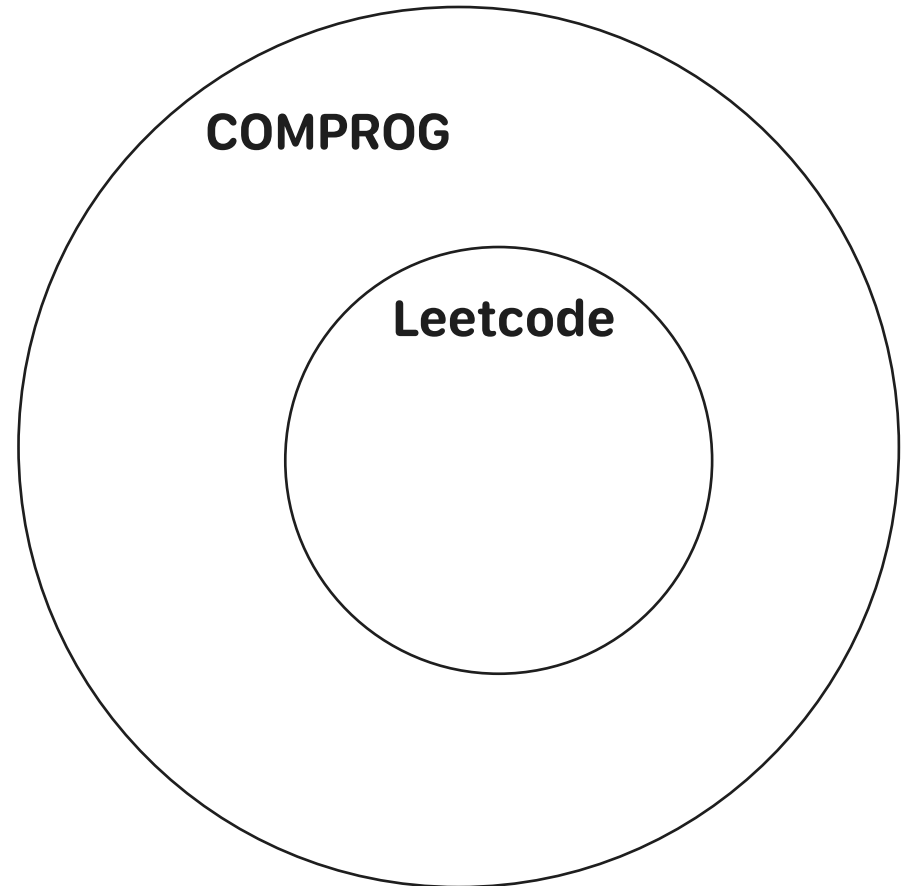


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November 21, 2025

Competitive Programming

- Race of solving logic/computational problems using algorithms.
- Coding interview but your direct candidates are UBC, Stanford olympiad undergrads



Interview-ish Problem

CSES Problem Set

Weird Algorithm

TASK STATISTICS

Time limit: 1.00 s Memory limit: 512 MB

Consider an algorithm that takes as input a positive integer n . If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for $n = 3$ is as follows:

$3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

Your task is to simulate the execution of the algorithm for a given value of n .

Input

The only input line contains an integer n .

Output

Print a line that contains all values of n during the algorithm.

Constraints

- $1 \leq n \leq 10^6$

Example

Input:

3

Output:

3 10 5 16 8 4 2 1

ICPC Problem



ICPC International Collegiate Programming Contest

The 2025 ICPC World Finals Baku

31 August – 5 September 2025 // hosted by ADA University

Problem A A-Skew-ed Reasoning

Time limit: 2 seconds

The following is based on a true story – the names have been changed because... well, because you always change names in stories like this one.

Professor Taylor Swift is grading a homework assignment on integer skew heaps. A skew heap is a binary tree with an integer stored in each node such that the value in any node is less than or equal to the values in any of its children. Note that the skew heap need not be a perfect binary tree; that is, the left and/or right subtree of any node may be empty.

Inserting a value x into a skew heap H is done using the following recursive procedure:

- If H is empty, make H a skew heap consisting of a single node containing x .
- Otherwise, let y be the value in the root of H .
 - If $y < x$, swap the two children of the root and recursively insert x into the new left subtree.
 - If $y \geq x$, create a new node with value x and make H the left subtree of this node.

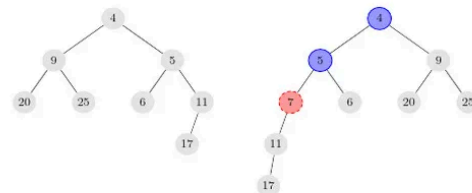
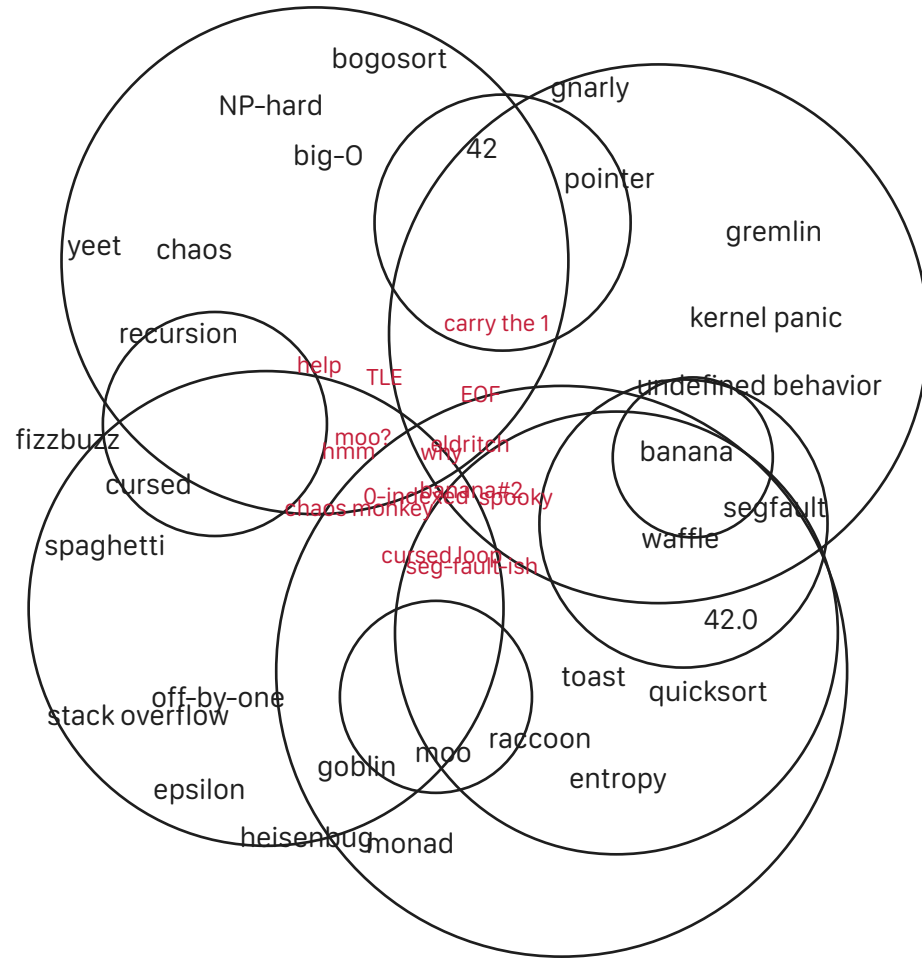


Figure A.1: Example of inserting the value 7 into a skew heap. The nodes storing 4 and 5 (marked in blue) have their children swapped, while the node storing 11 becomes the left child of the newly inserted node (marked in red).

Now, back to Professor Swift. The homework problem she has assigned asks the students to show the heap that results from inserting a given permutation of the numbers from 1 to n , in the given order, into an empty heap. Surprisingly, some of the students have wrong answers! That got Professor Swift wondering: For a given heap, is there an input permutation that would have produced this heap? And if so, what are the lexicographically minimal and maximal such input permutations?

Input

The first line of input contains an integer n ($1 \leq n \leq 2 \cdot 10^5$), the number of nodes in the tree. These nodes contain the numbers from 1 to n exactly. This is followed by n lines, the i^{th} of which contains two integers ℓ_i and r_i ($0 \leq \ell_i \leq n$ or $\ell_i = 0$; $0 \leq r_i \leq n$ or $r_i = 0$), describing the values of the left and right children of the node storing i , where a value of 0 is used to indicate that the corresponding child does not exist. It is guaranteed that this data describes a binary tree.



Accurate Venn diagram of programming

C++ Input / Output

- Standard input/output streams: `cin`, `cout`
- Basic usage:

```
int n;  
cin >> n;  
cout << n << "\n";
```

- Use `"\n"` instead of `endl` (avoids flushing the buffer)

Input

7

Output

7

Variables and Types

- Common types: `int`, `long`, `long long`, `double`, `char`, `bool`, `string`
- Watch out for overflow: use `long long` for large sums/products
- Declaring variables:

```
int a = 5;  
long long b = 1e18;  
double c = 3.14;  
string s = "hello";
```

- `auto` lets the compiler infer the type

Conditionals

- Conditionals:

```
if (x > 0) {  
    cout << "positive\n";  
} else if (x < 0) {  
    cout << "negative\n";  
} else {  
    cout << "zero\n";  
}
```

Input (x = -3)

Output

negative

For-loops

- For-loops:

```
for (int i = 0; i < n; i++) {  
    cout << i << " ";  
}
```

Input (n = 5)

Output

0 1 2 3 4

Dynamic Arrays (std::vector)

- std::vector resizes automatically, unlike a static array

```
vector<int> v(n);  
v.push_back(10);
```

- Common operations:

```
v.size();  
v.back();  
v.pop_back();  
sort(v.begin(), v.end());
```

- Access elements with v[i], just like a regular array

Problem 1

Problem Statement

- Alice wants to steal vacuum machines from n houses. The price of the vacuum machine in house i is $\text{price}[i]$.
- Read n , then the n prices.
- Print the total value Alice can steal (sum of all the prices).

Input

```
5
1 2 3 4 5
```

Output

```
15
```

Problem 1 - Solution

```
int n;  
cin >> n;  
vector<int> a(n);  
for (int i = 0; i < n; i++) cin >> a[i];  
  
long long sum = 0;  
for (int i = 0; i < n; i++) sum += a[i];  
  
cout << sum << "\n";
```